

# Buck Converter Application Design Calculations

Datasheet: <https://www.ti.com/lit/ds/symlink/tps54821.pdf>

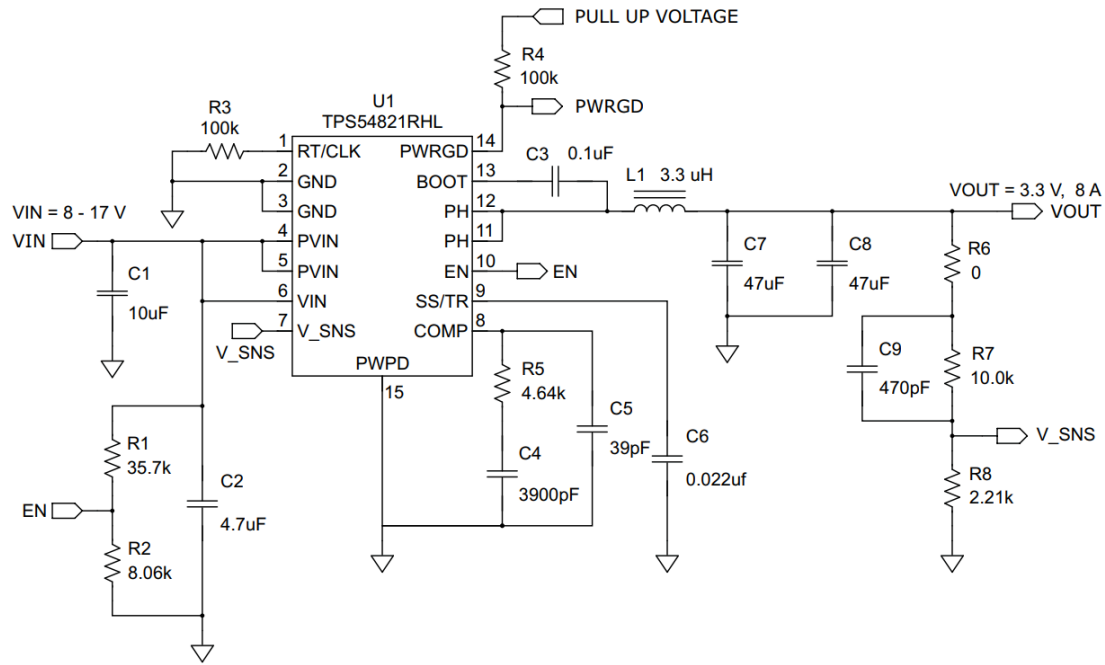


Figure 29. Typical Application Circuit

| PARAMETER                                             | VALUE                                                   |
|-------------------------------------------------------|---------------------------------------------------------|
| Output Voltage (Vout)                                 | 5 V                                                     |
| Output Current                                        | 8 A                                                     |
| Transient Response 4 A load step ( $\Delta V_{out}$ ) | $\Delta V_{out} = 0.35 \text{ V (7\%)}$                 |
| Input Voltage (Vin)                                   | 6 V (empty)<br>7.4 V (nominal)<br>8.4 V (fully charged) |
| Reference Voltage                                     | 0.6 V                                                   |
| Output Voltage Ripple                                 | 33 mV p-p                                               |
| Start Input Voltage (Rising Vin)                      | 6.528 V                                                 |
| Stop Input Voltage (Falling Vin)                      | 6.193 V                                                 |
| Operating Frequency                                   | 480 kHz                                                 |

## Operating Frequency

Switching frequency trade-off:

- Higher frequency = smaller physical solution size by allowing use of lower inductance values and output capacitors
- Lower frequency = more efficient and less heat

Source: <https://www.ti.com/lit/wp/slyy193c/slyy193c.pdf?ts=1780636757120>

A moderate switching frequency of 480 kHz was selected to balance compact solution size with high efficiency operation (same as application example).

## Undervoltage Lockout

Important for our 2S LiPo battery because over-discharging leads to permanent capacity loss of the cells:

Turn-On Threshold = 6.528 V

- Buck converter will not start switching until VBAT rises above this value
- Protects the converter from trying to run when the battery is too low

Turn-Off Threshold = 6.193 V

- Once operating, the buck converter will keep switching until VBAT falls below this value
- Shuts off below threshold to avoid undervoltage stress

Voltage Hysteresis (UVLO window)

- The difference in values prevents the converter from rapidly turning on/off if the battery voltage is hovering around the threshold
- Source: <https://www.ti.com/document-viewer/lit/html/SSZTAG8>

UVOL window is determined by the R1/R2 resistor divider feeding the EN pin

- R1 = 35.7 k $\Omega$ , R2 = 8.06 k $\Omega$  (same as application example)

## Bootstrap Capacitor Selection

A 0.1  $\mu$ F ceramic capacitor (C3) must be connected between the BOOT to PH pin for proper operation.

It is recommended to use a ceramic capacitor with X5R or better grade dielectric.

## Slow Start Capacitor Selection

Determines the minimum amount of time it takes for the output voltage to reach its nominal programmed value during power up

For our application, the soft start time is not critical since the output capacitor value is 2 x 47 uF which does not require much current to charge up to 5 V.

$$C6(nF) = \frac{T_{ss}(ms) \cdot I_{ss}(\mu A)}{V_{ref}(V)} \quad (I_{ss} = 2.3 \mu A \text{ for TPS54821})$$

Setting the soft start time to an arbitrary value of 6 ms, C6 = 22 nF (same as application example)

### Output Inductor Selection

KIND is coefficient that represents the amount of inductor ripple current relative to the maximum output current. Typical values for application range from 0.1 to 0.3. A value of 0.3 was chosen for this design.

$$L1 = \frac{V_{inmax} - V_{out}}{I_o \cdot K_{ind}} \cdot \frac{V_{out}}{V_{inmax} \cdot f_{sw}}$$

Minimum inductor value calculated as L1 = 1.76 uH

For this design, a larger standard value of 3.3 uH was chosen to achieve a lower ripple current and match the application example.

Source: <https://www.ti.com/lit/ta/ssztap6/ssztap6.pdf>

### Output Capacitor Selection

Response to large change in load current

- The regulator usually needs 2 or more clock cycles for the control loop to detect a change in load current and output voltage and adjust the duty cycle to accordingly
- The output capacitance must be large enough to supply the difference in current for 2 clock cycles while only allowing a tolerable amount of output voltage sag

$$C_o > \frac{2 \cdot \Delta I_{out}}{f_{sw} \cdot \Delta V_{out}}$$

Minimum capacitance value calculated as Co = 47.6 uF

To account for capacitance de-ratings due to aging, temperature, and DC bias, a large conservative capacitance value of 2 x 47 uF was chosen.

## Input Capacitor Selection

The TPS54821 requires a high quality ceramic, type X5R or X7R, input decoupling capacitor of at least 4.7  $\mu\text{F}$  of effective capacitance on the PVIN input voltage pins and 4.7  $\mu\text{F}$  on the VIN input voltage pin.

C1 = 10  $\mu\text{F}$  and C2 = 4.7  $\mu\text{F}$  capacitors were selected (same as the application example)

## Output Voltage Feedback Resistor Selection

The resistor divider network R7 and R8 is used to set the output voltage

$$R8 = \frac{R7 \cdot V_{REF}}{V_{OUT} - V_{REF}}$$

Setting R7 = 10 k $\Omega$ , R8 is calculated as 1.36 k $\Omega$ , rounded to R8 = 1.37 k $\Omega$  since that is the nearest standard 1% resistor

## Compensation Network

An RC network is used to stabilize the power converter's internal error amplifier and feedback control loop.

Compensation Resistor ( $R_C = R5 = 4.64 \text{ k}\Omega$ )

- Connects in series with the compensation capacitor ( $C_Z$ ) to the COMP pin
- Sets the mid-band gain and determines the loop crossover frequency (bandwidth) and transient response time

Compensation Capacitor ( $C_C = C4 = 3900\text{pF}$ )

- Creates a low-frequency pole, boosting the phase margin at the crossover frequency for loop stability

High-Frequency Capacitor ( $C_P = C5 = 39\text{pF}$ )

- Connects in parallel from the COMP pin to ground
- Creates a high-frequency pole to attenuate switching noise and high-frequency gain spikes

Feed-forward Capacitor ( $C_{FF} = C9 = 470 \text{ pF}$ )

- Connects in parallel to the top feedback resistor for  $V_{OUT}$
- Creates a phase-lead zero that increases phase margin, speeds up transient response, and helps attenuate high-frequency output noise

Because the operating conditions are similar to the application example, the same compensation values were used.